TECHNICAL WHITE PAPER

Bounded Perpetuals

Physical-First Solvency and Tranched Risk Capital for Oracle-Settled Markets

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How bounding an oracle-settled perpetual's payoff converts open-ended market risk into reservable credit risk - and enables physical-first solvency, tranched liquidity, and fail-soft settlement.

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Abstract

Perpetual markets normally manage an instrument whose price domain and cumulative settlement path are not bounded in advance. Plether Perps instead offers indefinite, oracle-settled directional exposure to a normalized six-currency basket whose mark is constrained to the interval *[0,C]*. The bounded price domain makes every position's price-PnL finite and permits a constant-time, common-mark liability envelope computed from two side aggregates. Plether combines that envelope with a USDC balance sheet that distinguishes physical cash from mathematical claims, a senior/junior LP capital stack, delayed historical-oracle execution, utilization-priced LP capital, and fail-soft terminal settlement. LP entry and exit use one exact signed mark-to-close adjustment: an Engine-maintained terminal book aggregates account-local price PnL from whole lots, exact entry cost, and collectible collateral caps.

The bound is useful but narrower than a blanket solvency guarantee. It limits gross positive **price PnL at one common mark**. It does not fully reserve every possible sequence in which Long and Short positions close at different prices, and it does not independently cap VPI rebates, carry, fees, oracle failures, or stablecoin losses. Plether therefore treats the endpoint envelope as a risk-admission and withdrawal primitive, not as a substitute for settlement containment. Profitable exits may become senior trader claims when cash is unavailable; valid terminal exits can complete even when they reveal insolvency; and a degraded-mode latch then prevents further risk expansion.

This paper formalizes the bounded-liability result and its counterexample, defines Plether's four accounting views, explains the HousePool capital waterfall and execution state machine, and evaluates the design against 2,685 ECB daily reference-rate observations from January 2016 through June 2026. A companion Python model reproduces selected Solidity accounting kernels, executes the numerical settlement vectors, and produces all reported empirical results from a deterministic scenario replay.

Status and scope

This paper describes the Plether Perps implementation at Git revision `06d0ab451ad9bb42f4e9869fc94b0eeb1e88efe5`. The implementation and accounting specification are normative where this paper and implementation differ. The paper is a market-design and accounting analysis, not investment advice, a promise of solvency, or a security audit. At this revision the system is pre-deployment. It has completed an external pre-audit consultation but not a formal production audit.

Contents

1. The market-design question

The modern crypto perpetual swap typically anchors a non-expiring contract to a spot index through transfers between long and short positions [1-3]. Other on-chain designs place a liquidity pool behind oracle-settled trader PnL or simulate execution against a virtual AMM. These approaches differ in execution and counterparty structure, but they share a difficult balance-sheet question: how much capital is enough when a reference asset can move without a contractual upper bound?

Plether starts from a different instrument:

> *Can an on-chain market provide leveraged, indefinite synthetic exposure without requiring balanced open interest, forced AMM selling, or treating unrealized trader losses as spendable assets?*

The answer developed here has five parts:

1. Bound the reference-price domain so price-PnL liability is finite.
2. Admit new risk against a constant-time endpoint envelope.
3. Keep physical cash, trader claims, reserved custody, terminal deficits, and LP equity as separate accounting objects.
4. Price directional concentration and the use of LP capital separately.
5. Preserve terminal settlement liveness even when it exposes a balance-sheet deficit.

Plether is therefore not best understood as a conventional matched order-book future. It is a pooled, oracle-settled perpetual CFD with USDC cross-margin custody, explicit position-margin buckets, and a tranched external capital provider. Long means long Plether's dollar index and gains as the raw FX-basket mark falls; Short means short the dollar index and gains as that mark rises. Accounts may hold one live side at a time, and changing direction requires closing first. The contract uses the same Long/Short product terminology even though its stored raw mark moves inversely to dollar strength.

1.1 Contributions and prior art

No individual mechanism in Plether should be presented as unprecedented. Capped, fully escrowed decentralized derivatives predate this work [4]. Pooled-counterparty perps, asynchronous post-commit oracle execution, integrated skew charges, utilization fees, and epoch-based LP entry all have direct precedents [13-19].

The design contribution is the combination:

- a capped price-PnL instrument;
- endpoint-derived, constant-time side liabilities;
- exact symmetric LP share pricing from an account-capped terminal NAV book;
- physical-first LP and solvency accounting;
- a senior/junior underwriting waterfall;
- LP-capital carry paid independently by both sides;
- senior trader claims and degraded containment rather than automatic profitable position deleveraging; and
- invariant-oriented separation of trader custody, queued intent, engine accounting, and pool capital.

This combination makes a precise claim possible: Plether can reserve the maximum gross positive price PnL of the current book at a common mark without iterating positions. It intentionally does **not** claim that the same reserve funds every future settlement path.

2. Instrument and basket

2.1 A normalized dollar-directional basket

Plether's mark is a normalized linear combination of six foreign currencies against USD:

Component	Weight	Deployment base USD price
EUR	57.6%	1.1750

Component	Weight	Deployment base USD price
JPY	13.6%	0.00638
GBP	11.9%	1.3448
CAD	9.1%	0.7288
SEK	4.2%	0.1086
CHF	3.6%	1.2610

For component USD prices $x_{j,t}$, deployment bases $x_{j,0}$, and weights w_j summing to one, the unconstrained basket is

$$\tilde{p}_t = \Sigma_j\, w_j (x_{j,t})/(x_{j,0}).$$

The engine mark used for PnL is

$$p_t = min(\tilde{p}_t, C),\ 0 \le p_t \le C.$$

The reference deployment uses $C=2$. The basket rises when its foreign currencies strengthen against USD and falls when USD strengthens.

The weights resemble those of the conventional ICE U.S. Dollar Index, but the instruments are not the same. ICE USDX is a proprietary geometrically weighted index with its own formula and market conventions [20]. Plether uses a normalized **linear** basket and must be described as dollar-directional or DXY-oriented, not as an on-chain DXY print.

2.2 Position price PnL

Let:

- $q_i \ge 0$ be position size;
- e_i *in* $[0,C]$ be its capped entry price;
- p *in* $[0,C]$ be a common settlement mark; and
- $[z]^+=max(z,0)$.

Signed price PnL is

$$\Pi_i^{LONG}(p)=q_i(e_i-p)$$

for Long and

$$\Pi_i^{SHORT}(p)=q_i(p-e_i)$$

for Short. The pool's gross positive price-PnL obligation at p is

$$G(p)= \Sigma_{i\ in\ LONG} q_i [e_i-p]^+ + \Sigma_{i\ in\ SHORT} q_i [p-e_i]^+.$$

Position margin is held in the MarginClearinghouse. Returning that trader-owned margin is distinct from paying positive price PnL from HousePool capital.

Proposition 1 - Common-mark endpoint envelope

Define the side endpoint liabilities

$$L_{LONG}=\Sigma_{i\ in\ LONG} q_i\, e_i$$

and

$$L_{SHORT}=\Sigma_{i\ in\ SHORT} q_i (C-e_i).$$

Then, for every common mark p *in* $[0,C]$,

$$G(p) \le L_{max} = max(L_{SHORT}, L_{LONG}).$$

Proof. Every term $[e_i-p]^+$ and $[p-e_i]^+$ is convex in p, so their nonnegative weighted sum $G(p)$ is convex. A convex function on a closed interval reaches its maximum at an endpoint. At $p=0$, Short positive PnL is zero and $G(0)=L_{LONG}$. At $p=C$, Long positive PnL is zero and $G(C)=L_{SHORT}$. Therefore the maximum is the larger endpoint value. *QED*

The engine stores side open interest, entry notional, margin, borrow base, and maximum-profit totals. A new position adds either $q\ e$ or $q(C-e)$ to one side. Computing L_{max} therefore requires two aggregate reads and one comparison, independent of account count.

The implementation fixes position size to whole 100-token lots. With an eight-decimal price, one lot times one price is already an exact six-decimal USDC atom quantity. For each account the Engine therefore stores the lot count and exact entry cost

$$E_i = \Sigma_j\ ell_{i,j} p_{i,j},$$

rather than reconstructing economic basis from a rounded average entry price. The average remains a display field only. Opens add exact endpoint profit, while a partial close assigns a proportional floor of entry cost to the closed lots and leaves every remainder atom on the surviving position. Exact entry cost and endpoint liability are consequently conserved across increases and partial closes. The earlier weighted-average dust counterexample is excluded by the canonical lot boundary; non-quantized open, increase, and close intents are rejected before entering the router queue.

2.3 What the envelope does not prove

Proposition 1 is an instantaneous result under one shared mark. It is not pathwise.

Consider one unit Long and one unit Short, both entered at $e=1$, with $C=2$. Each endpoint liability is 1, so $L_{max}=1$. At any one mark, aggregate positive price PnL is at most 1. But if Long closes at $p=0$ and is credited 1, then Short remains open and later closes at $p=2$, cumulative realized price-PnL obligations equal 2. The second obligation may be a trader claim rather than an immediate cash payment. Reserving 1 at inception did not fund this sequential path.

This counterexample is not an implementation accident. It identifies the exact boundary of the accounting primitive:

- `max(Long, Short)` is a common-mark price-PnL envelope;
- `Long + Short` is a more conservative bound on independently realized endpoint maxima;
- neither expression alone captures VPI, fees, carry, collateral seizure, claims, cash timing, or external loss; and
- the live protocol recomputes its balance sheet after each transition and may enter degraded mode after a terminal action.

Likewise, an individual **cash settlement** is not capped solely by its stored maximum price profit. A skew-reducing close can receive a negative VPI amount, while a position can also carry a negative lifetime VPI balance only when its gross clawback target is held in a dedicated reserve. The price-PnL envelope, VPI reserve, and all-in settlement equation must not be conflated.

3. A physical-first balance sheet

The central accounting discipline is to avoid using one quantity called "equity" for every decision. A protocol can be solvent for current risk admission while having no safely withdrawable LP cash. A trader can own a valid claim without the corresponding USDC being immediately available. A losing trader can owe value that has not entered HousePool custody.

3.1 Asset and obligation categories

Quantity	Meaning	Canonical location
Raw assets	Literal USDC token balance at HousePool	USDC `balanceOf(HousePool)`
Accounted assets	USDC admitted into protocol economics	HousePool accounting ledger
Physical assets A	Conservative backing `min(raw, accounted)`	`HousePool.totalAssets()`
Excess assets	Unsolicited raw surplus not yet assigned	`raw - accounted`, floored at zero
Free trader settlement	Withdrawable trader cash	MarginClearinghouse account
Position margin	Trader value locked behind a live position	Clearinghouse position bucket plus engine mirror
Order reservation	Margin committed to one queued order	Clearinghouse reservation keyed by order ID
Keeper reserve	Trader-funded order-execution bounty	Clearinghouse reserved-settlement bucket
Trader claims K	Senior pool obligations from deferred payouts	Engine beneficiary balances
Terminal deficit	Negative terminal LP equity at the authenticated mark	Engine/HousePool snapshot; blocks deposit activation
Unrealized trader gain	Exact marked LP obligation before close	Account terminal curve; not yet a cash claim
Unrealized trader loss	Account-capped marked LP receivable before close	Account terminal curve; never spendable pool cash
LP principals	Senior and junior claimant allocations	HousePool waterfall state
LP accounting equity	Physical backing net of claims plus the exact signed terminal price delta	HousePool reconciliation view, allocated by the waterfall
Treasury fees	Cash-realized protocol inventory	Treasury clearinghouse account

An unsolicited transfer to HousePool does not automatically increase economic depth. Conversely, a raw-balance shortfall reduces physical backing immediately. This `min(raw, accounted)` boundary prevents a donation from silently changing solvency, VPI depth, or share value before the transfer has an assigned owner.

Terminal NAV V2 has no accumulated-debt ledger or owner repayment selector. An unpaid trader payout remains explicit as a senior trader claim. Independently negative terminal LP equity is exposed as terminal deficit and blocks deposit activation. Price loss above an account's collectible cap was never included as an LP receivable and is emitted only as diagnostic write-off telemetry; a terminally uncollectible action-charge remainder is waived.

Marked ownership is sign-symmetric even though cash availability is not. An unrealized trader gain reduces LP share NAV. An unrealized trader loss increases that same NAV only up to value collectible from the losing account's dedicated PnL pledge and nettable same-account claim. The receivable remains non-cash and cannot fund a redemption until settlement collects it. Senior and junior principal allocate the resulting terminal equity through the waterfall. These categories must not be collapsed merely because an off-chain dashboard can net them into one number.

3.2 Four views for four questions

Accounting view	Question	Primary construction
Risk admission	May the protocol accept more price risk?	Post-operation physical assets net of trader claims versus L_{max}
LP withdrawal	How much physical USDC may leave now?	Physical assets less endpoint liability, trader claims, and explicit reserves
Tranche reconciliation and share pricing	What marked ownership backs tranche shares?	Physical assets less claims plus the exact signed account-capped terminal price delta
Terminal reachability	What trader-owned value can this close or liquidation actually consume?	Explicit clearinghouse buckets and permitted terminal reservation paths

The views share source data but answer different questions. Reusing a less conservative view merely because its field names look similar is an accounting error.

3.3 Risk-increasing admission

Let:

- A^+ be projected post-operation physical assets;
- K^+ be projected aggregate trader claims;
- P^+ be a pending payout not yet reflected in A^+; and
- L_{max}^+ be the projected common-mark endpoint envelope.

The engine's effective assets are

$$A_{eff}^+ = max(A^+ - K^+ - P^+, 0).$$

Risk expansion is rejected unless

$$A_{eff}^+ \geq L_{max}^+.$$

For a simple open with no preexisting claims or pending payout, this reduces to the familiar expression

$$A_{physical} \geq max(L_{SHORT}, L_{LONG}).$$

Open planning includes immediate pool changes. Positive VPI increases physical pool assets; a VPI rebate reduces them; the protocol execution fee belongs to treasury rather than LP assets. The admission test therefore uses the projected post-operation balance sheet rather than a stale pre-trade asset number.

Proposition 2 - Constant-time admission predicate

Given correctly maintained side maximum-profit aggregates and current physical/claim totals, evaluating the engine's risk-admission predicate is $O(1)$ in the number of positions.

This is a statement about computational complexity and conformance to the protocol's chosen predicate. Because of the sequential-settlement counterexample in Section 2.3, it is not a theorem that every future close path is fully funded.

3.4 LP withdrawal firewall

The base withdrawal reservation is

$$R_{base} = L_{max} + K + R_{sup},$$

where R_{sup} is the explicit supplemental-reserve slot. Live HousePool paths can layer pending claimant buckets and unassigned assets on top. The base free cash is

$$W_{free} = max(A - R_{base}, 0).$$

Proposition 3 - Snapshot withdrawal preservation

Assume the snapshot already satisfies $A \geq R_{base}$. If an LP withdrawal w satisfies $0 \leq w \leq W_{free}$, then the remaining physical assets satisfy

$$A - w \geq R_{base}.$$

The proof follows directly from the definition of W_{free} under the stated precondition. Equivalently, the live withdrawal gate permits success only from an already reserved snapshot and only up to its computed free cash. The property preserves the current snapshot reserve; it does not extend Proposition 1 into a pathwise guarantee.

Boundary example. Let physical assets be \$60 million, trader claims \$5 million, and L_{max} \$55 million. Effective assets equal \$55 million, so the risk balance is exactly at its chosen solvency boundary. Withdrawal reserves equal all \$60 million and LP free cash is zero. "Solvent" and "withdrawable" are therefore not synonyms.

3.5 Exact symmetric terminal NAV

Share pricing asks an ownership question, not a cash-withdrawal question. It therefore uses the same exact mark-to-close adjustment for incoming and outgoing LPs while retaining the endpoint envelope of Section 3.4 for physical redemption funding.

For account i, let ell_i be its number of 100-token lots, E_i its exact entry cost in USDC atoms, and k_i its effective collectible cap: the smaller of its dedicated PnL pledge plus nettable same-account claim and its maximum possible price loss inside $[0,C]$. With the protocol's LONG side profiting as the oracle price falls and SHORT profiting as it rises, the LP-side terminal price delta is

$$delta_i^{LONG}(p) = min(ell_i\, p - E_i, k_i)$$

and

$$delta_i^{SHORT}(p) = min(E_i - ell_i\, p, k_i).$$

A negative value is marked value owed by LPs to that trader. A positive value is a trader loss recognized for LP ownership only to the extent the same account can pay it. Carry, VPI, fees, frozen spreads, liquidation reserves, order margin, and action reserves are separate realized-settlement quantities and are excluded from this curve.

`TerminalNavBookV2` maintains

$$\Delta_{terminal}(p) = \Sigma_i delta_i(p)$$

without iterating active accounts. Each account contributes a piecewise-affine curve with at most one cap breakpoint. A fixed-depth radix-16 prefix tree over the 32-bit price domain aggregates its base coefficient and breakpoint event, so reads and mutations are bounded independently of account count. Only the immutably bound Engine can set or remove a curve. Domain-separated curve hashes authenticate replacement, and a monotonic book version exposes every effective mutation.

At the Engine's authenticated cached mark, the common share-pricing snapshot is

$$T = A - K + \Delta_{terminal}(p),\ D_{reconcile} = max(T,0),\ D_{deficit} = max(-T,0),$$

before any additional explicit claimant or unassigned-asset adjustments. The Engine updates position lots, exact entry cost, pledge/claim state, and the account curve atomically; HousePool separately enforces mark-age and frozen market policy before LP actions.

Both deposit activation and redemption share pricing use this one signed snapshot. A new entrant therefore receives the same discount for inherited marked trader-profit liabilities borne by an exiting LP, and receives credit only for collectible trader losses. Positive marked receivables still do not increase the physical cash available to fund redemptions. A current terminal deficit blocks deposit activation rather than minting ownership into negative equity.

3.6 Conservation is category-specific

Let H be physical USDC in HousePool and Q physical USDC in the MarginClearinghouse across trader, treasury, and keeper ownership buckets. For one internal settlement transition with no deposit, withdrawal, unsolicited transfer, stablecoin rebase, or token failure:

$$\Delta(H+Q) = 0.$$

That token identity is necessary but incomplete. A settlement can replace a cash payout with a non-cash trader claim, consume an existing claim against a price loss, write off price loss that was never recognized beyond the account's collectible cap, or waive an uncollectible terminal action charge. Those items change entitlement or telemetry without minting physical USDC.

Proposition 4 - Transition-level conservation of value categories

Assume the planner and apply path use the same settlement amounts and the underlying USDC transfers conserve tokens. For every successful close or liquidation:

1. every physical debit has one physical credit across HousePool, MarginClearinghouse ownership buckets, and any explicit external flow;
2. a deferred fresh payout increases trader claims by exactly the unpaid payout and causes no contemporaneous HousePool cash debit;
3. claim consumption reduces the same beneficiary liability and offsets only that account's exact realized price loss;
4. price loss above the precommitted collectible cap is emitted as a diagnostic write-off and creates no asset, claim, deficit, or protocol debt;
5. every surviving account curve equals its remaining lots, entry cost, and post-settlement collectible cap; and
6. for a frozen voluntary close,

$$spread_{assessed} = spread_{paid} + spread_{waived}.$$

Proof sketch. The price-GAIN branch either transfers the complete fresh payout or creates an equal claim; it never does both. The price-LOSS branch partitions exact realized loss into same-account claim consumption, PnL-pledge seizure, and an explicit amount above the cap that was absent from pre-close LP NAV. Carry, VPI, execution fees, frozen spread, and liquidation charges use separate action and liquidation reserves; a terminally uncollectible action remainder is waived rather than recast as price debt. The apply path then installs the residual curve or removes the full-close curve atomically. Liquidation separately partitions its dedicated charge reserve among keeper, protocol, and LP recipients. Each branch is a disjoint exhaustive partition. The proposition proves ledger conservation conditional on correct planning and application; it does not prove that the resulting claims are liquid or that the pool remains solvent.

The executable vectors test these identities directly. The on-chain evidence is the economic/value-conservation invariant family discussed in Section 11.

4. HousePool as a capital structure

LPs are not passive AMM depositors in this design. They underwrite the bounded price-PnL claims of the trader book and receive realized trading revenue, carry, and frozen-market compensation. Their capital is divided into senior and junior ERC-4626 vaults [8].

4.1 Senior and junior claims

Let S and J denote senior and junior principal. Let H be the senior high-water mark: the unimpaired senior entitlement that future revenue must restore before junior receives residual value.

The waterfall applies these rules:

1. Loss reduces junior principal first.
2. Loss beyond junior reduces senior principal.
3. Loss does not reduce H.
4. Later revenue first restores S to H.
5. Revenue remaining after senior restoration accrues to junior.
6. The senior target coupon is paid from junior principal and is capped by available junior principal.
7. Coupon paid above an unimpaired senior balance ratchets both S and H, protecting that paid coupon from later junior extraction.

Prioritized claims can transform the distribution of risk but cannot remove aggregate risk. Structured-finance literature is especially clear that senior safety depends on the quality of the underlying loss model and on systematic tail dependence [10-11]. Plether's tranche labels are priorities, not credit ratings.

4.2 Waterfall example

Suppose the pool begins with:

- senior principal *S=$60* million;
- junior principal *J=$40* million; and
- senior high-water mark *H=$60* million.

A $50 million reconciliation loss produces:

- *J=0*;
- *S=$50* million; and
- *H=$60* million.

If $25 million of revenue later becomes distributable, the first $10 million restores senior to its high-water mark and the remaining $15 million establishes junior principal. The resulting state is *S=60*, *J=15*, *H=60*, all in millions of USDC.

4.3 Synchronized LP epochs are an accounting and liquidity control

Ordinary entry and exit are fully asynchronous. Both tranches share a round-hour clock and one permissionless Router coordinator backed by a HousePool callback that is Router-only while live positions exist outside oracle-frozen mode. Request-epoch selection is symmetric across Senior and Junior and across deposits and redemptions. For inclusion time *t*, let *e=lfloor t/3600rfloor* and *b=(e+1)times3600*. Every request targets *e+1* when *t<b-300*, and *e+2* when *t≥ b-300*. Exact cutoff equality therefore rolls forward rather than reverting. The target remains numerically continuous across the round-hour boundary, and its start is always more than five minutes and at most sixty-five minutes after inclusion.

The five-minute interval is a maximum-membership quiet period for the epoch about to mature, not a price commitment or protocol freeze. Cancellations may reduce that epoch, while trading, claims, oracle updates, and pool accounting remain live. Settlement maturity remains `currentEpoch >= requestId`, and the live-position reconciliation mark is still required to be published at or after the round-hour boundary rather than the request cutoff. One bounded settlement transaction then:

1. validates a pool-reconciliation Pyth basket, installs the exact Engine mark, and reconciles HousePool accounting from one signed terminal-NAV and waterfall snapshot;
2. funds matured senior redemption demand before any junior redemption demand;
3. funds junior demand only from remaining free liquidity and subordinated capital capacity;
4. finalizes matured junior deposits and then matured senior deposits from the same signed accounting snapshot; and
5. leaves funded assets or shares in vault escrow for later user claims.

Redemption priority is demand-based: dormant senior NAV does not reserve cash from junior claimants, but any newly matured senior request moves ahead of an older unfunded junior remainder. Incoming deposits never expand the withdrawal budget captured for the same settlement call. A complete pending deposit is cancellable before maturity. After maturity, cancellation follows the documented epoch rejection, terminal-wipe, impairment, and Senior-capacity escape policy. A complete pending redemption is cancellable only before maturity while it is wholly unfunded and no refund has activated. Funded claims remain callable independently of later settlement, pause, or oracle liveness. If no queued epoch can advance, settlement reverts and rolls back its reconcile and carry checkpoints together with the Pyth and Engine mark updates; this prevents permissionless no-op calls from changing time-based accounting. With live open positions, the basket's earliest publish time must be at or after the round-hour boundary. Cached-mark settlement is retained only for zero-position state and the explicit frozen-oracle exit regime.

This implements the pending-to-claimable lifecycle standardized for asynchronous vaults in ERC-7540 [9], with protocol-specific shared request ids, terminal refund states, and bounded queue rules.

The delay separates request funding from activation; it is not a substitute for exact valuation. At settlement, both deposit activation and redemption pricing consume the current signed TerminalNavBookV2 adjustment. A close or liquidation that converts marked price PnL into physical cash, a trader claim, or an explicit write-off also replaces or removes the account curve in the same transition. Finalizing before or after that transition therefore cannot select the retired deposit-only valuation: each valid snapshot applies one economic state to both LP directions.

Epochs do not remove ordinary timing and liveness risk. There is no auction or user-specified share-price limit, so a matured request is priced at the valid snapshot when a permissionless caller actually settles it. Until activation, deposit assets remain in request escrow under the cancellation policy rather than silently becoming active LP capital.

4.4 Frozen-market LP operations

When the oracle is genuinely frozen, new deposit requests are unavailable and queued deposits remain escrowed rather than activating against a frozen mark. Eligible exits remain live under a tranche-local surcharge. Reference defaults are 25 basis points for senior and 75 basis points for junior. At synchronized settlement, redemptions pay fewer net assets against the gross tranche claim. The fee remains in the same tranche and does not become protocol revenue. Preview methods revert for these asynchronous flows; explicit exit estimate views expose nonbinding current quotes.

4.5 Recapitalization and ownership routing

"More USDC in the pool" is not one accounting event. Plether distinguishes:

- **Claimant recapitalization with cash arrived:** new cash intended to restore the LP waterfall;
- **Revenue with cash arrived:** a fresh claimant-owned inflow, such as collected trading revenue;
- **Revenue already retained:** cash is already physical in HousePool and only its ownership is being assigned, so accounted assets must not increase twice;
- **Excess admission:** governance explicitly admits an unsolicited raw-token surplus into canonical assets.

Revenue restores impaired senior principal to its high-water mark before junior participates. Recapitalization can restore claimant value and solvency capacity, but it does not automatically clear degraded mode: the explicit owner recovery action remains necessary after the balance sheet is genuinely restored.

5. Pricing capital and directional concentration

Plether separates five prices that are often collapsed into a generic "trading fee."

Charge or adjustment	Economic purpose	Primary beneficiary
Execution fee	Protocol service revenue on traded notional	Treasury, only when cash credited
VPI	Price a change in directional skew	Pool when positive; trader rebate when negative
Carry	Rent for LP capital committed behind maximum-profit exposure	LP claimants when realized
Oracle confidence shift	Side-adverse execution within Pyth uncertainty	Protective execution adjustment
Frozen close spread	Compensate LPs for voluntary stale-mark exit	LPs only; never treasury

5.1 Virtual price impact

At fixed pool depth D, Plether assigns absolute skew S the potential

$$C(S)=(kS^2)/(2D),$$

where k is the VPI factor. A trade pays

$$VPI = C(S_{post}) - C(S_{pre}).$$

A trade increasing absolute imbalance pays a positive amount; a trade reducing imbalance earns a negative amount. Because VPI is the difference of a state potential, a sequence at unchanged D telescopes:

$$\Sigma_n \Delta C_n = C(S_N) - C(S_0).$$

Splitting the same skew change into pieces therefore does not avoid VPI under fixed depth. The implemented integer potential also telescopes exactly when each charge is the difference of the same floor-rounded potential at unchanged depth; flooring changes its value relative to the continuous curve but not that algebraic cancellation. Changing depth between trades breaks the fixed-potential assumption, and the position lifetime clamp can truncate close rebates. Partial-close VPI release is a bounded linear approximation rather than a fresh exact curve evaluation. Liquidation computes no fresh VPI delta, though negative accrued VPI is included in its action-charge settlement.

The gross target $max(-VPI_{accrued},0)$ is held as a dedicated nonwithdrawable sub-balance of action reserve. An open or increase that makes the target larger must fund it immediately or fail. Generic action collection cannot cross the combined floor protecting execution bounties and this VPI reserve. VPI does not add to or subtract from P+C price-risk equity: reserve value above the target never adds price collateral, while reserve value below the target is an independent delinquency that blocks withdrawal and makes the account liquidatable. A close or liquidation consumes the reserve when the clawback is realized and releases only value no longer required by the surviving target. A new close-time action rebate still comes only from pool cash free of existing trader claims; an unfunded remainder is waived rather than claim-backed. The reserve is excluded from the terminal price curve, so LPs cannot withdraw it as marked equity before the matching position obligation settles.

Integrated skew pricing has clear prior art in Synthetix and Perennial [14,18]; the relevant Plether contribution is how this pricing interacts with the bounded balance sheet and lifetime accounting.

Example. With $D=\$100$ million, $k=0.005$, and skew moving from \$10 million to \$30 million:

$$C(10m)=\$2,500,\ C(30m)=\$22,500.$$

The VPI charge is \$20,000. Reversing the move at unchanged depth produces a \$20,000 rebate before the lifetime clamp.

5.2 LP-capital carry

For position i, Plether defines an LP-backed borrow base

$$b_i=max(L_i^{max}-m_i,0),$$

where m_i is active position margin and L_i^{max} is that position's maximum price profit. Subtracting margin is a **carry-pricing convention**: it prices only the portion of maximum-profit exposure governance treats as rented LP capacity. Trader-owned position margin remains in MarginClearinghouse and does not reduce HousePool's endpoint reserve. A fully margined position may therefore have zero borrow base while its full maximum price profit remains in the HousePool liability aggregate.

For side s, the implementation's zero-asset edge cases are explicit:

$$B_s=\Sigma_{i\ in\ s}b_i,\ u_s=min(B_s/A,1)\ when\ A>0.$$

If $B_s=0$, utilization is zero. If $B_s>0$ while $A=0$, utilization is defined as 100% rather than dividing by zero.

If r_{max} is the annual simple carry rate at full utilization, the side rate is

$$r_s=r_{max}u_s.$$

Each position accrues against a cumulative side index:

$$carry_i=b_i\Delta I_s.$$

The implementation performs the utilization and time multiplication in one combined integer expression to retain more precision than first rounding a displayed basis-point rate.

This is capital rent, not a transfer from the heavier side to the lighter side. Both Long and Short can pay simultaneously because both can reserve LP balance sheet capacity simultaneously. Carry continues during stale and frozen oracle windows. Health checks first project it against eligible free settlement. A fully funded carry obligation therefore does not reduce the separate exact price-risk health basis; any uncovered remainder blocks withdrawal and independently makes the account liquidatable. PnL pledge plus same-account claim backs only exact price risk and cannot offset that remainder. Before a mutation changes its basis or rate denominator, elapsed carry is checkpointed. The protocol collects it physically when that is safe; otherwise the amount remains in the account-specific `unsettledCarryUsdc` bucket for later collection. A checkpoint therefore preserves accrued history without pretending every checkpoint is an immediate cash realization.

Example. Let pool assets be \$100 million and the full-utilization base rate be 5%:

Side	Max-profit envelope	Position margin	Borrow base	Utilization	Effective annual rate	One-year carry
Long	\$40.0m	\$4.0m	\$36.0m	36.0%	1.800%	\$648,000

Side	Max-profit envelope	Position margin	Borrow base	Utilization	Effective annual rate	One-year carry
Short	\$25.0m	\$2.5m	\$22.5m	22.5%	1.125%	\$253,125

The example produces \$901,125 of total one-year carry before changes in positions, pool depth, or collection. The Solidity rate-view helper displays the Short rate as 112 basis points after flooring, while the combined index calculation preserves the 112.5-basis-point economic product.

6. Delayed execution and oracle policy

Plether does not expose a same-transaction trader market-order path. A user commits a binding intent; a keeper later executes the global FIFO head against verified Pyth data. Delayed settlement is a recognized defense against oracle latency exploitation, while finite settlement windows limit the option value given to the trader [5,15]. Delay does not eliminate transaction-ordering MEV, and FIFO does not by itself prove fairness [21-22].

6.1 Order lifecycle

1. The trader deposits USDC into MarginClearinghouse.
2. The trader constructs one typed request with an account-scoped nonzero `clientOrderId`, side, size, margin, nonzero price limit, close/open intent, deadline, permitted execution modes, expected execution-configuration hash, and inclusive financial limits.
3. `commitOrder` permanently binds `(account, clientOrderId)` to the full intent hash in OrderLifecycleBook. An exact replay returns the original order ID before current-state validation and creates no second side effect; conflicting reuse reverts.
4. For a fresh valid intent, the router records commit time and block, moves open margin into an order-specific reservation, and reserves a keeper bounty from trader value.
5. The binding, user-uncancellable order becomes the global FIFO head in turn.
6. For live execution, a keeper supplies Pyth updates and native-token oracle fees.
7. The router resolves the unique historical update in the configured window strictly after commit.
8. Basket components are normalized, confidence is propagated, publish-time dispersion is checked, and the side-adverse execution price is capped at *C*.
9. The stateless policy evaluator rebuilds authoritative Engine state, derives protocol meaning through the canonical planner, and checks the resulting economics against the request's pinned modes and bounds.
10. Exact known planner or policy failures may terminate; timing, close-only, insufficient-gas, mark-ordering, unknown, panic, out-of-gas, empty, or malformed failures leave the FIFO head pending for retry.
11. Each prepared batch item executes in an independent Router rollback frame, so one retryable item cannot undo an already finalized prefix. Oracle/Pyth preparation remains batch-atomic.
12. A terminal item settles reserved value exactly once, deletes ephemeral Router state, and then finalizes an authenticated lifecycle receipt. The Book retains the compact outcome and receipt hash permanently while emitting the full receipt preimage for independent reconstruction of what happened.

Pyth's `parsePriceFeedUpdatesUnique` primitive verifies that returned updates are the first qualifying feed updates in a specified range [6]. Plether obtains "strictly after commit" by setting the lower bound after the commit timestamp, not because Pyth globally assigns that semantic. Pyth confidence is an uncertainty interval width, not a confidence score [5,23].

6.2 Basket-specific oracle checks

The PletherOracle:

- fixes feed IDs, weights, bases, inversion flags, and Pyth endpoint per oracle deployment;
- converts each feed to the shared price scale;
- computes the weighted normalized basket;
- propagates basket confidence from weighted component relative confidence;

- rejects a component whose confidence-to-price ratio exceeds policy;
- rejects excessive publish-time dispersion between basket legs;
- uses the oldest component publish time as the basket timestamp; and
- caps the basket and execution price at *C*.

Live and FAD-only trades use a side-adverse fraction of the Pyth confidence interval. Oracle-frozen voluntary closes instead use the unshifted validated basket and pay a fixed LP-owned spread. Liquidations retain conservative side-adverse pricing and do not pay the frozen voluntary-close spread.

6.3 Regimes as a product state machine

Oracle condition and balance-sheet condition are separate dimensions. Define the execution state

$$M=(O,D),\ O \text{ in } Live,FAD,Frozen,\ OverStale,\ D \text{ in } Healthy,Degraded.$$

The oracle/calendar component *O* moves according to the FAD schedule, frozen calendar, authenticated publish times, and configured freshness limits. It returns toward `Live` only when those exogenous conditions again pass. The orthogonal latch *D* moves from `Healthy` to `Degraded` when a permitted terminal close or liquidation leaves projected physical assets net of claims below the current admission envelope. It does not auto-clear: after genuine recapitalization or risk reduction restores the balance sheet, an explicit owner action is required to return it to `Healthy`.

An action is authorized only if both coordinates permit it. Thus a healthy but over-stale market still blocks settlement, while a degraded but live market permits protective closes and liquidations but no new position risk. The table below is the resulting permission projection; `Degraded` intersects every oracle row rather than replacing it.

Regime	Risk increase	Voluntary close	Liquidation	Principal policy
Live	Allowed if all checks pass	Live historical tick	Live conservative tick	Normal margin and freshness
FAD/runway	Blocked	Live close-only tick	Available	Elevated FAD margin
Oracle frozen, within limit	Blocked	Validated stale-window price plus fixed spread	Available under frozen policy	Risk reduction prioritized
Frozen beyond max staleness	Blocked	Blocked	Blocked	Wait for valid data
Degraded	Blocked	Allowed when oracle regime permits	Allowed when oracle regime permits	Protective transitions and recapitalization

FAD addresses scheduled FX-market closure risk. BIS guidance notes that major FX markets trade through the working week but not over weekends, so the reopening rate may jump relative to Friday [24]. Plether's reference defaults start a close-only runway before the closure period and raise the margin basis to 3%. Administrators may add expected FX-holiday dates.

6.4 Keeper economics and queue liveness

Keeper rewards are reserved from trader clearinghouse value, not drawn from HousePool. Open and close orders use free settlement. Close commitment may checkpoint carry first, but it never reclassifies PnL pledge to fund a bounty, including on the stale-mark path. Execution-time user failures and specified protocol-state invalidations pay the keeper so an invalid FIFO head can be removed.

Binding orders reduce the ability to treat a queue position as a free option, but they remove user cancellation after an order has entered the queue. The global head also creates a liveness dependency: no keeper may skip it merely because a later order is more attractive. Order age, per-account pending limits, bounded cleanup, typed failure categories, and reserved bounties are the compensating controls.

7. Settlement and failure containment

The protocol prioritizes completing valid **terminal** risk-reducing transitions over preserving the appearance of pre-transition solvency. A partial close may write off price loss above its precommitted collectible cap, but it still reverts if a separate action charge must be collected in full or if the surviving position and terminal curve cannot remain consistent.

7.1 Profitable close

A close first computes:

$$\textit{net settlement} = \textit{realized price PnL} - \textit{VPI} - \textit{execution fee} - \textit{frozen spread} - \textit{pending carry}.$$

When this value is positive, the pool pays it immediately only if physical cash remaining after reserving **existing aggregate trader claims** covers the entire fresh payout. Otherwise the complete fresh amount becomes a new beneficiary-based trader claim. There is no partial immediate payout and no FIFO claim queue.

Claims are senior in:

- risk-admission effective assets;
- LP withdrawal reserves;
- tranche reconciliation; and
- future cash-priority decisions.

An account claim may be settled only by its beneficiary and only when aggregate claims are fully covered. Settlement credits the MarginClearinghouse account rather than transferring directly to the wallet.

Claim example. Suppose HousePool has $18 million of physical assets and $15 million of existing claims. Only $3 million is free for a fresh payout under the claim-priority rule. A profitable full close owed $5 million completes, but the full $5 million becomes a new claim. The mechanism preserves position-exit liveness; it does not provide immediate liquidity.

7.2 Losing partial and full closes

For a losing close, the exact entry cost allocated to the closed lots determines price loss. Settlement nets a same-account trader claim first and then consumes the dedicated PnL pledge. It does not treat order margin, liquidation reserve, execution-bounty reserve, or generic action reserve as additional price-loss backing. Any price loss beyond the pre-close account cap is a diagnostic write-off because LP NAV never counted it as receivable.

A partial close is therefore not blocked merely because uncapped price loss exceeds collectible value. Instead it must:

1. consume the exact claim and pledge amounts selected by the planner;
2. retain any pledge required to make the residual curve conserve the pre-close terminal value;
3. collect required carry, VPI, execution-fee, and frozen-spread action charges in full; and
4. atomically replace the account curve with the remaining lots, entry cost, and collectible cap.

A full close follows the same separated price channel, may waive a terminally uncollectible action-charge remainder, and removes the curve. Price loss above the cap never creates protocol debt or a terminal deficit.

Underwater close example. Suppose an exact closed-lot basis produces an $800,000 trader price loss while the account has no nettable claim and only $300,000 of PnL pledge. LP NAV immediately before the close includes a $300,000 marked receivable, not $800,000. Settlement transfers $300,000 to HousePool, emits a $500,000 price-loss write-off, and removes or replaces the account curve. The recognized terminal value is conserved. Execution fee, carry, VPI, and any frozen spread are assessed through their separate action-charge path.

7.3 Liquidation

Liquidation eligibility uses exact P+C price-risk health. Negative lifetime VPI is a separate typed obligation: reserve backing below its gross target is an independent liquidation condition, while excess reserve never adds price collateral. Carry is likewise isolated: eligible free settlement covers pending carry first, while any uncovered carry independently makes the account liquidatable and cannot be offset by PnL pledge or claim. A liquidation is a full-position terminal transition; there is no partial-liquidation recovery path for an oversized account. Liquidation then:

1. consumes exact price loss from same-account claim and PnL pledge and writes off only the excess;
2. assesses the capped liquidation charge against its dedicated reserve and allocates the configured keeper and protocol shares plus the exact LP remainder;
3. settles carry and the negative lifetime-VPI clawback through their separate action path, waiving any terminally uncollectible remainder;

4. preserves a positive fresh trader payout as cash or a senior trader claim;
5. removes the position, terminal curve, and bounded account-local pending order state; and
6. evaluates degraded mode.

Liquidation allocation example. If the capped liquidation charge is $2,000 and its dedicated reserve contains $2,000, the keeper and protocol allocations round down from that amount and LPs receive the exact remainder. Price PnL is settled independently against the account's claim and PnL pledge; an uncovered price-loss tail neither reduces the $2,000 allocation nor creates protocol debt.

Because liquidation settles against an external oracle mark, one liquidation does not mechanically trade through an AMM curve and move the next account's execution price. A common oracle shock can still liquidate many accounts at once, and keeper/oracle capacity remains a liveness dependency.

Plether uses claims and degraded containment rather than automatically reducing profitable positions solely to repair a pool PnL ratio. This differs from oracle-pool designs that employ auto-deleveraging as a solvency backstop [17].

7.4 Frozen voluntary closes

During oracle freeze, the reference configuration assesses a fixed 50-basis point spread on reduced notional. The spread:

- replaces the live side-adverse confidence shift for voluntary closes;
- belongs entirely to LP claimants;
- never credits treasury;
- remains separate from signed VPI;
- must be fully collectible for a partial close; and
- may be partly waived only for a terminal full close.

For every valid frozen voluntary close,

$$spread_{assessed} = spread_{paid} + spread_{waived}.$$

Liquidations never assess this spread.

Frozen terminal-exit example. A full close reduces $2 million of notional, so the 50-basis-point frozen spread is $10,000 and the 4-basis-point execution fee is $800. Suppose the position also has a $2,000 price loss, no VPI or pending carry, and only $5,000 is reachable. The $12,800 obligation is allocated in priority order:

- $800 pays the execution fee;
- $2,000 pays the entire base price loss;
- $2,200 pays part of the LP-owned frozen spread;
- $7,800 of spread is waived; and
- no protocol liability or terminal deficit is created because all base loss was collected.

The full close completes and *$10,000=$2,200+$7,800*. A partial close with the same shortfall would revert.

7.5 Degraded mode

After a close or liquidation, the engine compares effective physical assets with the remaining endpoint envelope. If the former is smaller, degraded mode latches. While degraded:

- new opens are blocked;
- position-backed withdrawals are blocked;
- closes and liquidations remain available when oracle policy permits;
- mark updates remain available; and
- recapitalization can restore the balance sheet.

Degraded mode is not a global pause. It is a latched containment response to a terminal transition that the protocol deliberately allowed to finish; it does not auto-clear merely because a later mark looks solvent. After genuine restoration, the owner may clear the latch through the explicit recovery path.

8. Empirical evaluation

8.1 Data and method

The empirical dataset contains 2,685 complete ECB daily reference-rate observations from 4 January 2016 through 30 June 2026 [7]. ECB quotes each currency as units per EUR. For non-EUR component j, the model derives

$$x_{j/USD} = (x_{USD/EUR})/(x_{j/EUR}),$$

then applies Plether's deployment weights and bases.

The fixed API response has SHA-256:

`45a8b4e376c3f518c6687335ba849d087cb34906e1bfa6217e0305e0223b4eca`

ECB rates are published on working days for information purposes and are not recommended as transaction prices [7]. The study is therefore a transparent macro-price proxy, not a replay of Pyth ticks. The descriptive statistics do not model confidence width, intraday paths, keeper latency, or tradable Friday-close/Monday-open quotes. Section 8.4 adds a stateful daily-fix scenario replay with integer PnL, admission, carry, fees, liquidation, claims, and legacy uncovered-loss telemetry, and two explicitly non-production tranche overlays: a realized-cash view and a conservative side-envelope stress after each observation's ordinary actions, plus one final reconciliation after forced end-of-sample closes. The replay predates the exact account-capped TerminalNavBookV2 kernel and does not estimate production LP share prices. It also holds VPI, oracle confidence, frozen execution, slippage rejection, pending epochs, and intraday keeper behavior outside the experiment.

Plether basket history

8.2 Basket range and cap distance

Statistic	Result
Complete observations	2,685
Mean normalized basket	1.0070
Minimum	0.8636 on 28 Sep 2022
Maximum	1.1168 on 15 Feb 2018
Maximum as fraction of cap $C=2$	55.84%
Largest peak-to-trough decline	22.67%
Last observation	0.9695 on 30 Jun 2026

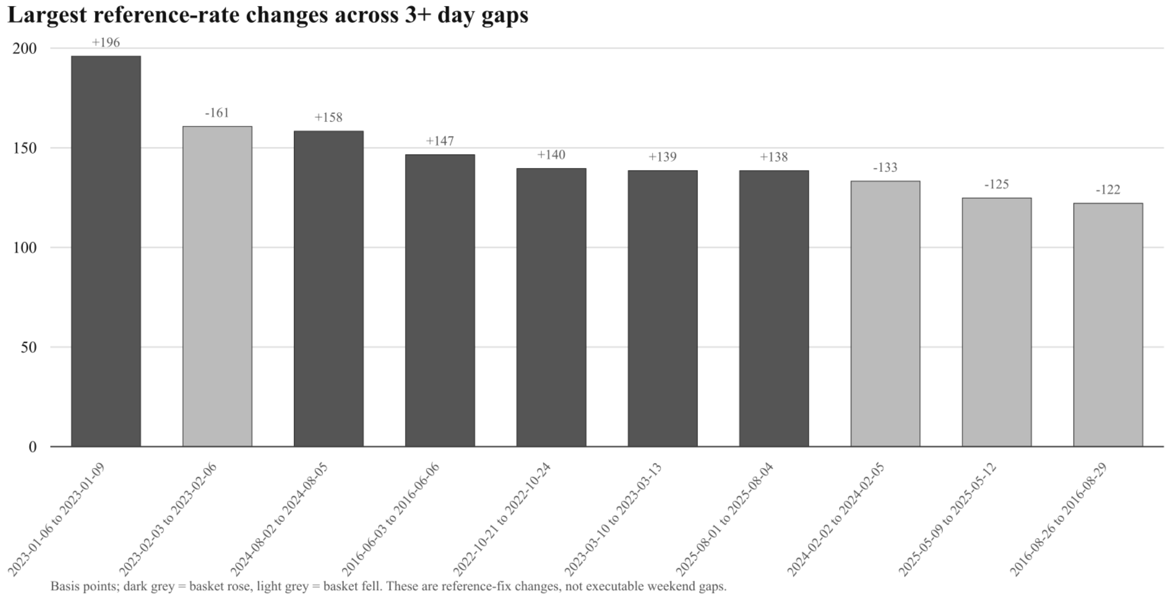
The cap did not bind in this sample. That is not evidence that it cannot bind. It shows the capital-efficiency price of an endpoint guarantee: a large fraction of reserved tail space was unused by the historical macro proxy.

For the most favorable forward Short move in the sample, a position entering at 0.9597 and exiting at a later 1.1168 used 15.09% of its cap-endpoint maximum-profit reserve. For Long, the highest reserve utilization was 22.67%, from an entry at 1.1168 to a later 0.8636. These are ex-post sample maxima, not risk limits.

8.3 Adjacent fixes and weekend proxy

Interval sample	Count	Median absolute change	95th percentile	99th percentile	Maximum
All adjacent observations	2,684	22.67 bps	83.03 bps	136.10 bps	358.37 bps
Gaps of at least 3 calendar days	550	24.53 bps	92.11 bps	138.54 bps	195.97 bps
Friday-to-Monday observations	523	24.61 bps	90.33 bps	138.54 bps	195.97 bps

The largest Friday-to-Monday reference-fix move was +195.97 basis points from 6 January to 9 January 2023. The largest adjacent-fix move overall was +358.37 basis points from 10 to 11 November 2022.



Largest nonpublication gap changes

As a deliberately simple move-threshold diagnostic, the model counts absolute reference-rate changes greater than 1%, 1.5%, and 3%:

Absolute move threshold	All adjacent intervals	3+ day gaps	Friday-Monday
1.0%	79	22	20
1.5%	18	3	3
3.0%	1	0	0

No Friday-to-Monday reference-fix change in the sample exceeded 3%. This does **not** validate a 3% FAD margin basis as sufficient. These thresholds are not liquidation buffers: actual eligibility depends on starting equity relative to the applicable maintenance basis, current notional, carry, lifetime VPI clawback, confidence adjustment, and execution timing. The calculation also ignores the tradable path between fixes and extreme events outside the sample.

8.4 Stateful historical cohort replay

The replay starts with $100 million of physical HousePool cash split into $50 million senior and $50 million junior principal. That LP capital is static: there are no deposits, withdrawals, or recapitalizations during a scenario. At the first available fix of each month, after a 20-observation signal history:

1. it requests a $90 million entry-notional cohort;
2. the allocation signal uses the previous fix versus the fix 20 observations before it;
3. the current fix is the one-observation-lagged entry-price proxy;
4. the signal-aligned side receives 50%, 80%, or 100% of requested notional and the other side receives the remainder;
5. the paired allocation is scaled on an integer 10^{18} grid until both endpoint solvency and the reference 40% final-skew wall pass;
6. each position posts 1.5% margin, has a 1% maintenance requirement, and is scheduled to close after 63 calendar days;
7. carry accrues at 5% annually at full side utilization;
8. the senior coupon checkpoints at the 8% reference rate over each observed calendar interval;
9. voluntary closes pay the 4-basis-point treasury execution fee; and
10. each intervening ECB observation tests liquidation with the reference 10-basis-point bounty and $1 floor.

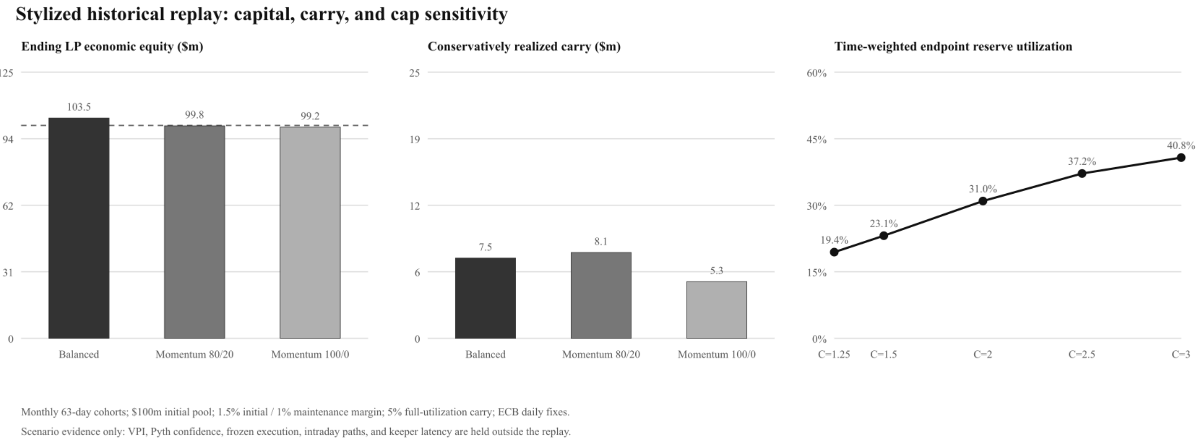
Each position uses a unique synthetic account whose terminal reachable collateral is only its posted position margin; no additional free or committed cross-margin collateral is supplied. The replay holds maintenance at 1% throughout and does not switch to the 3% FAD basis around closure windows. The paired cohort is an aggregate research allocation whose legs are assumed to be interleaved; the model checks final skew but does not simulate transaction-level intermediate skew, VPI, or queue execution.

The historical model labels uncollectible price loss as uncovered-loss telemetry. That label is not current Terminal NAV V2 settlement semantics: production caps the marked receivable by same-account claim plus PnL pledge and emits any excess as `PriceLossWrittenOff` without creating protocol debt or an LP deficit. The reported telemetry below is retained only to reproduce the published research run.

After price and parameter ratios are quantized, cohort scaling and every accounting transition use integer arithmetic. A post-construction assertion rechecks endpoint solvency and skew for every admitted cohort; there is no arbitrary minimum scale. The baseline processes liquidations before scheduled closes and uses ascending account order inside each class. Claims can be serviced only at a strictly later observation where physical cash covers the entire aggregate, and every position still open at the end of the sample is forcibly full-closed at the last fix. Opening fees are funded by the synthetic trader account rather than HousePool.

Metric	Balanced 50/50	Momentum 80/20	Momentum 100/0
Admitted / scaled / rejected cohorts	64 / 0 / 61	93 / 85 / 32	105 / 105 / 20
Cumulative admitted entry notional	$5.760bn	$5.759bn	$3.904bn
Time-weighted endpoint reserve utilization	30.21%	30.95%	22.20%
Time-weighted capped aggregate borrow-to-assets	32.68%	32.50%	21.94%
Time-weighted Long / Short carry utilization	17.04% / 15.64%	15.80% / 16.79%	10.84% / 11.16%
Liquidations	100	145	87
Liquidations after a 3+ day gap	25	38	18
Carry assessed	$7.600m	$8.100m	$5.340m
Carry conservatively realized	$7.531m	$8.060m	$5.319m
Legacy loss telemetry	$0.991m	$0.919m	$0.654m
Trader claims created	$0	$0	$0
Ending LP economic equity	$103.493m	$99.817m	$99.196m

There are 125 monthly requests in each scenario. In the balanced case, later one-sided liquidation can leave a skewed live book that an equal-size paired cohort cannot heal; 61 such requests have no admissible positive scale. The 80/20 and 100/0 allocations hit the hard-skew constraint in 85 and 105 admitted cohorts respectively. This is why the directional scenarios admit less notional than a model that silently disables the skew wall.



Historical replay summary

The liquidation counts remain intentionally sobering. At 1.5% initial margin and 1% maintenance, an account begins with only about a 0.5%-of-notional price-loss buffer before carry and other adjustments. Testing only at daily fixes almost certainly understates intraday liquidations while sometimes overstating the loss realized at a delayed daily mark.

Ordering sensitivity did not change this daily-grid 80/20 result. All four combinations of liquidation-first versus scheduled-close-first and ascending versus descending account order produced 145 liquidations, 40 scheduled closes, $0.919 million of legacy uncovered-loss telemetry, zero claims, $99.817 million of ending LP equity, and a $22.944 million minimum senior principal in the legacy side-envelope stress shadow. That equality is a result of these observations and event dates, not a theorem that settlement ordering is immaterial.

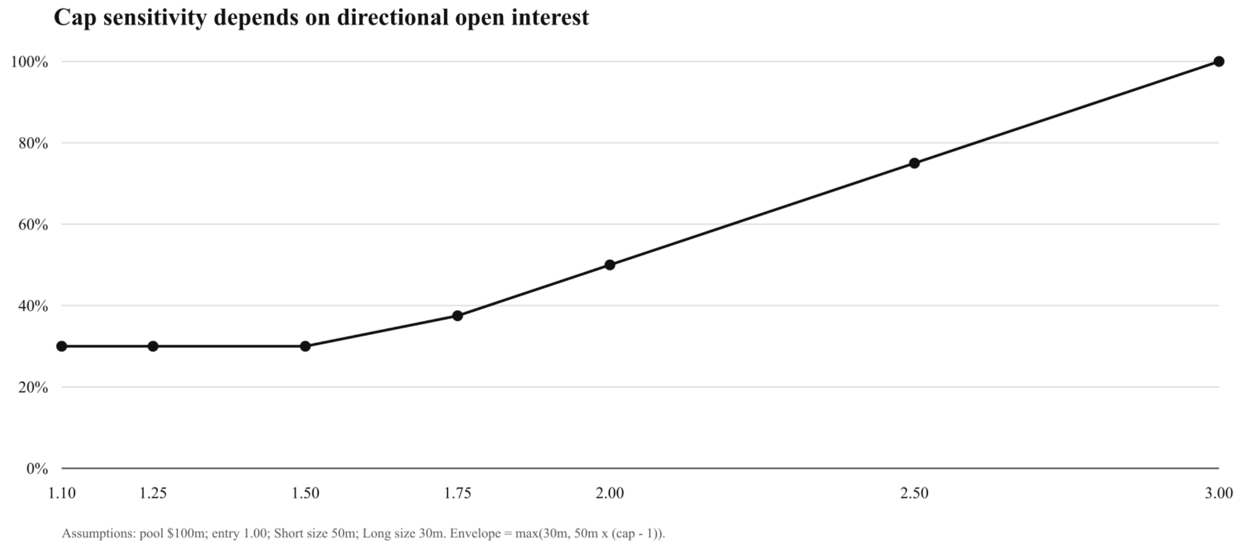
Each scenario has one forced end-of-sample close. These ending values are scenario outputs, not profitability forecasts. The momentum allocation is a deterministic book generator, not a claim that momentum is profitable or representative of production demand.

8.5 Cap and directional-book sensitivity

Consider a stylized $100 million pool, entry price 1.00, 50 million Short units, and 30 million Long units. The common-mark endpoint envelope is

$$L_{max}(C)=max(30m,50m(C-1)).$$

Cap	Long endpoint	Short endpoint	L_{max}	Pool reserve utilization
1.10	$30.0m	$5.0m	$30.0m	30.0%
1.25	$30.0m	$12.5m	$30.0m	30.0%
1.50	$30.0m	$25.0m	$30.0m	30.0%
1.75	$30.0m	$37.5m	$37.5m	37.5%
2.00	$30.0m	$50.0m	$50.0m	50.0%
2.50	$30.0m	$75.0m	$75.0m	75.0%
3.00	$30.0m	$100.0m	$100.0m	100.0%



Cap sensitivity

The result is piecewise. Lowering the cap below 1.60 does not reduce this book's reserve because Long remains the dominant endpoint. Cap governance cannot be analyzed without side composition.

Re-running the stateful 80/20 scenario changes both admitted size and the carry base:

Cap	Cumulative admitted notional	Time-weighted reserve utilization	Realized carry	Ending LP equity	Legacy loss telemetry
1.25	$5.601bn	19.43%	$4.117m	$97.126m	$0.904m
1.50	$5.591bn	23.13%	$4.879m	$97.123m	$0.905m
2.00	$5.759bn	30.95%	$8.060m	$99.817m	$0.919m
2.50	$5.891bn	37.16%	$12.495m	$107.043m	$0.949m
3.00	$5.702bn	40.75%	$15.850m	$110.259m	$1.023m

Admitted notional is non-monotonic in this path. A larger cap raises Short endpoint liability and borrow base, but the 40% skew wall, prior liquidations, cash evolution, and the admission state jointly determine which later cohorts fit. Carry rises across these particular runs because the surviving admitted book rents more maximum-profit-minus-margin capacity. Neither pattern is a general theorem about volume or LP returns. The $C=2.5$ and $C=3$ runs also triggered and, under the stated research assumption, later owner-cleared degraded mode twice and four times respectively.

8.6 Claims and tranche loss distribution

No historical replay produced a trader claim. This zero was **not imposed**: the cash-or-claim branch ran on every positive settlement, after reserving outstanding claims. In these scenarios, no fresh payout exceeded available physical cash. Claim duration is therefore undefined rather than reported as zero. Because the scenarios hold LP capital static, this result is conditional on the absence of LP withdrawals and other external HousePool cash outflows.

This does not make claims irrelevant. They become active after cash is reserved for existing claims, after physical impairment, or when sequential settlements realize the path risk in Section 2.3. The $18m cash / $15m claims / $5m fresh payout vector executes the same kernel and creates a $5m claim:

$$\textit{new claim if fresh payout} > A - K.$$

Tranche results require a valuation convention. The first view below is a **realized-cash overlay**: it applies each change in physical cash minus claims through the waterfall and checkpoints the 8% senior coupon over each observed interval. It omits the signed terminal price delta and therefore is not the production share-NAV view.

Realized-cash overlay metric	Balanced 50/50	Momentum 80/20	Momentum 100/0
Cumulative senior coupon transfer	$54.447m	$52.046m	$50.566m
Realized loss-event count	27	40	18

Realized-cash overlay metric	Balanced 50/50	Momentum 80/20	Momentum 100/0
Median realized loss event	$1.002m	$0.591m	$1.335m
95th-percentile realized loss event	$2.439m	$2.549m	$2.791m
Maximum realized loss event	$2.611m	$3.458m	$2.791m
Minimum junior principal after coupon and settlement	$0	$0	$0
Minimum senior principal	$50.000m	$50.000m	$50.000m

Junior reaches zero partly because it funds the senior coupon; that is not the same category as a trading loss. Losses can still impair senior relative to its coupon-ratcheted high-water mark even when its absolute principal never falls below the initial $50 million.

The second view is a **daily conservative side-envelope stress shadow** retained from the original empirical study. After each ECB observation's actions, it computes

$$D_{shadow} = max(A-K-M(p),0)$$

and reconciles the coupon-adjusted waterfall to that value. This is not the production kernel: $M(p)$ is the retired side-aggregate envelope and ignores both account entry distributions and collectible caps. The current protocol instead uses $\Delta_{terminal}(p)$ from Section 3.5. The legacy shadow remains useful only as an intentionally conservative sensitivity view. Its automatic once-per-observation cadence is also a research convention; the replay performs one additional reconciliation at the final fix after forcibly closing every remaining position.

Legacy side-envelope stress metric	Balanced 50/50	Momentum 80/20	Momentum 100/0
Maximum side-envelope stress	$50.424m	$78.368m	$66.418m
Minimum distributable value	$50.813m	$22.944m	$36.854m
Minimum junior principal	$0	$0	$0
Minimum senior principal	$50.000m	$22.944m	$36.854m

The shadow view reverses any inference that the replay proves senior safety. In the directional scenarios, this deliberately conservative stress can materially impair senior even though realized HousePool cash remains near $100 million. Because it is not the exact account-capped production NAV, it does not quantify current Senior or Junior share value; it demonstrates sensitivity to valuation assumptions.

As a separate counterfactual gross price-PnL stress with the skew gate disabled, applying the largest observed Friday-to-Monday change to a fully utilized one-sided $100m book produces a $1.960m loss: junior falls to $48.040m and senior remains $50m. A hypothetical $60m shock wipes junior and impairs senior to $40m while the high-water mark remains $50m. A 100% one-sided book is not admissible under the reference 40% skew wall. This overlay also ignores trader margin recovery and all charges.

8.7 Interpretation

The empirical results support five modest conclusions:

1. The cap creates a tail reserve far beyond moves in this ten-year daily proxy, which improves boundedness but consumes capital.
2. Weekend closure deserves a distinct policy even though the historical Friday-Monday fixes were smaller than the largest weekday change.
3. Default 1.5% initial margin produces many daily-fix liquidation events once the 1% maintenance requirement and carry are applied.
4. Claim formation is state- and sequence-dependent; its absence in these historical scenarios is evidence about the scenarios, not a protocol guarantee.
5. Realized cash and the legacy side-envelope stress can diverge sharply; the replay does not substitute for an account-level TerminalNavBookV2 simulation of production share NAV.

The dataset does not establish future loss probabilities, protocol profitability, safe leverage, or an optimal cap.

9. Comparison with other perpetual architectures

Dimension	Matched CLOB perpetual	vAMM perpetual	Unbounded oracle pool	Plether bounded oracle pool
Execution price	Order matching	Endogenous curve	External oracle plus impact rules	Historical external basket plus VPI/confidence policy
Counterparty	Matched traders/clearing system	Virtual inventory plus collateral vault	LP pool	Tranched USDC HousePool
Price domain	Generally unbounded	Generally unbounded	Generally unbounded	Contractually capped *[0,C]*
Recurring payment	Long-short funding	Funding and/or curve economics	Funding/borrow fees	LP-capital carry; both sides can pay
Concentration price	Book spread/depth	Curve slippage	OI impact/funding	Quadratic VPI plus hard skew limit
LP NAV treatment	Not applicable or insurance fund	Protocol-specific	Often includes trader PnL	Physical-first; exact signed account-capped terminal PnL, with marked receivables kept non-cash
Tail backstop	Insurance/social loss/ADL	Protocol-specific	OI caps, reserves, ADL	Endpoint admission, claims, degraded mode, recapitalization
Liquidation feedback	Book execution can move price	Curve trade can move price	Oracle-settled	Oracle-settled

Synthetix documents a pooled counterparty with skew-driven funding and OI caps [13-15]. GMX uses oracle-based pooled markets with reserve factors, adaptive funding/borrowing, LP valuation that includes pending trader PnL, and ADL in specified conditions [16-17]. Earlier Perpetual Protocol designs illustrate the vAMM alternative [19].

Plether's principal distinction is not that it has an oracle, a pool, delayed orders, or a skew curve. It is that a capped payoff permits endpoint-derived side liabilities, while physical-first accounting refuses to turn unsettled loser debt into LP cash. The cost is capital conservatism, asynchronous LP entry, and a settlement path that can produce explicit senior claims rather than pretend every valid close is immediately liquid.

This also differs from a balanced-OI funding model. In a matched long-short market, recurring funding is primarily a transfer between opposing trader cohorts and balanced open interest is the natural clearing condition. Plether does not require one side's notional to fund the other. The HousePool can underwrite net directional open interest up to its bounded-credit admission limits, and carry prices each side's use of LP-backed capacity independently. Consequently both sides can pay carry at once: their rents compensate pool capital rather than forcing a zero-sum funding transfer between Long and Short.

10. Security model and limitations

10.1 Assumptions

The strongest paper claims require:

- all executed marks are correctly authenticated, normalized, and capped;
- every live entry price lies in *[0,C]*;
- side aggregates remain equal to the underlying live-position state;
- HousePool `totalAssets()` correctly reflects conservative physical USDC;
- clearinghouse buckets preserve ownership and reachability semantics;
- privileged contracts and one-time dependency wiring are correct;
- USDC maintains the settlement value assumed by the model, remains transferable with standard ERC-20 behavior, and is not made unavailable by issuer blacklisting or an issuer-controlled upgrade;
- keepers can submit eligible orders and liquidations within policy windows; and
- governance acts within disclosed timelocks and parameter bounds.

The contracts are non-upgradeable. Risk parameters and oracle address rotation remain governance-controlled behind the documented delays, while core deployed logic and `CAP_PRICE` are immutable.

10.2 Oracle risk

Pyth or its publishers can be wrong, unavailable, stale, or internally dispersed. The basket can combine components whose economic liquidity differs. Exponent normalization and Solidity arithmetic round. Historical update data may be unavailable to keepers. Confidence is an uncertainty measure, not a guarantee that a true price lies inside the interval. General oracle literature also distinguishes source manipulation, aggregation, delivery, and consumer-contract failures [23].

Plether mitigates but does not eliminate these risks through:

- delayed unique historical settlement;
- side-adverse confidence use;
- component confidence and publish-time-dispersion limits;
- distinct freshness windows;
- close-only freeze policy;
- immutable cap; and
- timelocked oracle rotation.

10.3 Economic and accounting limits

- **Sequential settlement:** L_{max} is not a pathwise reserve.
- **Lot lattice:** exact entry-cost and terminal-book arithmetic relies on every open, increase, and close being a whole 100-token lot; changing the price decimals or quantum invalidates the USDC-atom identity and requires a new deployment.
- **Cap selection:** a tighter cap lowers some liabilities but changes the economic instrument and may not reduce the dominant side.
- **Capital efficiency:** historical moves can use only a small fraction of the endpoint reserve.
- **VPI scope:** the continuous and floor-rounded potentials telescope at constant depth, but changing depth and lifetime clamps alter path outcomes; partial-close release is a bounded linear approximation, and liquidation computes no fresh VPI.
- **Carry governance:** the rate curve is parameterized, not a market-clearing theorem.
- **Claims:** claims preserve exit-state liveness, not immediate redemption.
- **Price-loss tails:** loss above the account's precommitted collectible cap is written off rather than converted into LP equity, deficit, trader claim, or protocol debt; correctness depends on keeping that cap synchronized with pledge and nettable claim state.
- **Tranches:** junior subordination redistributes loss; it does not reduce total loss.
- **Terminal NAV:** exactness depends on atomic synchronization among position lots, entry cost, PnL pledge, same-account claims, and each account curve; deployment therefore binds one empty book to one Engine with no repair path.
- **Deposit epochs:** synchronized pricing removes the entry/exit NAV mismatch, and live marked finalization atomically validates a post-boundary oracle basket, but permissionless finalization still has timing and liveness risk because requests have neither an auction nor a user-specified share-price limit.
- **Frozen spreads:** a fixed spread does not scale with staleness inside the permitted window.
- **Frozen execution precision:** frozen-oracle exits deliberately sacrifice price precision for exit liveness by using the last validated stale-window basket plus a fixed spread.
- **Historical replay:** daily ECB fixes, synthetic monthly cohorts, isolated scenario accounts, paired/interleaved cohort admission, constant 1% maintenance, static LP capital, an observation-level reconciliation shadow plus a post-forced-close reconciliation, forced final closes, and zero VPI/confidence/frozen adjustments are research controls, not a production backtest.
- **Stablecoin risk:** accounting in USDC units does not guarantee one-dollar economic value, unrestricted transferability, standard token behavior, or freedom from issuer blacklisting and issuer upgrades.

10.4 Execution and liveness limits

- Global FIFO can be delayed by an unexecutable head until policy permits retry or finalization.
- Queued binding orders cannot be cancelled by users.
- Keepers require native-token fees, working infrastructure, and historical oracle payloads.
- Component publication schedules can diverge.
- Over-stale frozen conditions intentionally block even risk-reducing execution.
- A common market shock can make many accounts liquidatable simultaneously.
- Oracle settlement avoids AMM price-impact cascades but not congestion, correlated insolvency, or keeper scarcity.

10.5 Governance and implementation risk

Timelocks make changes observable; they do not make them correct. Senior coupon, margin ratios, VPI factor, skew cap, carry rate, freshness limits, keeper bounties, frozen spreads, and LP surcharges all embody economic judgment. Non-upgradeability reduces logic-replacement risk while increasing the cost of repairing an undiscovered flaw. Sidecars with settlement authority and any future external method added to them must be reviewed as core custody code.

11. Verification strategy

The white paper's Python model independently reproduces selected integer kernels for PnL, maximum profit, the legacy conservative side-envelope stress, VPI, carry indexes, cash-priority payouts, close-loss allocation, liquidation, solvency, and the tranche waterfall. It tests Proposition 1 over real-valued books and one-shot stored-integer positions, reproduces the historical non-quantized multi-increase rounding counterexample that motivated exact lots, executes the conservation vectors, includes the sequential-close counterexample, and runs the historical cohort scenarios. It does not implement the account curves or radix accumulator of TerminalNavBookV2 and is not a second implementation of the complete router, oracle, clearinghouse, or epoch state machines.

The direct Solidity unit, fuzz, matrix, and differential suites remain the highest-signal evidence for individual arithmetic kernels. Stateful evidence and known gaps are mapped to the invariant families documented in the invariant guide:

Claimed property	Invariant evidence or explicit gap
Capped PnL and endpoint arithmetic	Full-path conservation in `PerpExplicitAccountingInvariant.t.sol`; direct exact-lot arithmetic and close-conservation tests cover increases and partial closes, while non-quantized intents are rejected at router and Engine boundaries
Constant-time endpoint aggregation and admission arithmetic	Companion model and direct arithmetic/open-planning tests; no stateful invariant proves computational complexity
Exact account-capped terminal NAV and symmetric LP pricing	`TerminalNavBookV2.t.sol`, `TerminalNavCloseConservation.t.sol`, `TerminalNavIntegrationSecurity.t.sol`, `HousePool.t.sol`, and synchronized epoch integration tests; the Python replay does not reproduce the radix book
Protocol accounting-view alignment	`PerpPreviewInvariant.t.sol`, `PerpEconomicConservationInvariant.t.sol`
Withdrawal-reserve composition	`PerpEconomicConservationInvariant.t.sol`; Proposition 3 is the algebraic snapshot result
Preview/live close and liquidation parity	`PerpExplicitAccountingInvariant.t.sol`, `PerpClosePreviewParityInvariant.t.sol`, `PerpPreviewInvariant.t.sol`
Physical and ownership-category conservation	`PerpEconomicConservationInvariant.t.sol`, `PerpValueConservationInvariant.t.sol`, `PerpExplicitAccountingInvariant.t.sol`
Trader-claim totals, isolation, and cash gating	`PerpTraderClaimInvariant.t.sol`, `PerpMultiAccountInvariant.t.sol`, `PerpEconomicConservationInvariant.t.sol`
Account-capped price collection, write-off isolation, and failed-full-close value safety	`TerminalNavCloseConservation.t.sol`, `TerminalNavIntegrationSecurity.t.sol`, `PerpEconomicConservationInvariant.t.sol`, and `PerpValueConservationInvariant.t.sol`; close/liquidation cleanup and preview parity also use `PerpAccountingInvariant.t.sol` and `PerpExplicitAccountingInvariant.t.sol`

Claimed property	Invariant evidence or explicit gap
Universal terminal completion and new-risk blocking	Direct close/liquidation and degraded-mode policy tests; no stateful invariant spans every valid insolvent terminal path and every new-risk entry point
Degraded transition and preview/live parity	`PerpPreviewInvariant.t.sol`, `PerpExplicitAccountingInvariant.t.sol`
Carry history and ownership	`PerpValueConservationInvariant.t.sol`; utilization/rate arithmetic and simultaneous two-side carry are direct/model tests, not statefully invariant-tested
VPI lifetime conservation	No nonzero-VPI stateful invariant at this revision; direct VPI unit/fuzz/matrix tests are required
Frozen-spread conservation	No successful nonzero frozen-spread stateful invariant at this revision; dedicated frozen-close tests are required
Oracle/FAD/freshness boundaries	`PerpOracleBoundaryInvariant.t.sol`, `PerpOraclePathInvariant.t.sol`
Regime authorization matrix	Direct oracle-freshness/router/engine policy tests; boundary invariants do not span the complete two-axis action matrix
FIFO and reservation conservation	`PerpAccountingInvariant.t.sol`, `PerpMultiAccountInvariant.t.sol`
Binding order fields and first unique post-commit tick	Direct `OrderRouter.t.sol` tests; no dedicated stateful invariant covers intent immutability or strict historical-tick uniqueness
Active tranche lifecycle, cooldowns, and excess	`PerpHousePoolLifecycleInvariant.t.sol`, `PerpValueConservationInvariant.t.sol`
Junior-first loss, high-water restoration, coupon ratchet, and recapitalization priority	Direct `HousePool.t.sol` tests plus companion-model vectors; no dedicated stateful waterfall invariant at this revision
Synchronized LP deposit/redemption epochs	Dedicated cutoff, coordinator, FIFO, allocation-dust, plateau-liveness, and exact inverse-rounding integration/fuzz tests; `GovernedSeniorCapacityInvariant.t.sol` statefully covers shared cutoff routing across both request directions plus reservation/covenant safety, but not the complete settlement-phase/backlog state space
Account isolation	`PerpMultiAccountInvariant.t.sol`
Fee custody	`PerpFeeFlowInvariant.t.sol`

Tests demonstrate conformance within their harnesses; they are not formal proofs of the entire system. The invariant guide documents the suites, ghost ledgers, handlers, and these coverage boundaries.

12. Conclusion

Bounding a perpetual market's price domain does something valuable and precise: it turns each position's price-PnL into a finite contingent claim and permits a constant-time upper envelope for gross positive price PnL at a common mark. That is enough to build clearer admission, withdrawal, and LP-capital accounting than an unbounded pooled liability permits.

It is not enough to promise perpetual solvency. Traders settle over time, the mark moves between settlements, all-in cash flows include more than price PnL, and physical liquidity can fail independently of mathematical entitlement. Plether's serious design choice is therefore not the cap alone. It is the combination of a bounded snapshot envelope with accounting that admits what is physical, what is reserved, what is owed, and what has failed to arrive.

The HousePool makes that credit structure explicit. Junior and senior LPs own different loss priorities. Carry prices the LP capital actually supporting maximum-profit exposure. VPI prices directional concentration. Delayed execution prices oracle latency. Trader claims make cash shortfall visible. Degraded mode contains a deficit without trapping users in positions solely to preserve a pre-close accounting state.

The result is best characterized as **bounded-credit perpetual market infrastructure**: finite in instantaneous price-PnL geometry, explicit about temporal and liquidity risk, and designed so the balance sheet remains legible when normal settlement assumptions fail.

Appendix A. Notation

Symbol	Definition
A	Canonical physical HousePool assets
C	Immutable protocol price cap
D	Pool depth used by VPI
E_i	Exact entry cost of account i, in USDC atoms
e_i	Position entry price
$G(p)$	Gross positive price PnL at common mark p
H	Senior high-water mark
J	Junior principal
K	Aggregate trader-claim liability
k_i	Effective collectible price-loss cap of account i
ell_i	Number of canonical 100-token lots in account i
L_i^{max}	Position maximum price profit
L_{LONG}	Long endpoint liability at $p=0$
L_{SHORT}	Short endpoint liability at $p=C$
L_{max}	Larger side endpoint liability
$M(p)$	Legacy side-envelope stress used only in the empirical replay
m_i	Position margin
p	Capped basket mark
q_i	Position size
S	Absolute directional skew, or senior principal where stated
T	Signed terminal LP equity before the tranche waterfall
u_s	LP borrow-base utilization for side s
w_j	Basket component weight
$delta_i(p)$	Account i's signed LP-side terminal price delta
$\Delta_{terminal}(p)$	Exact aggregate signed terminal price delta

Appendix B. Reproduction

The companion source is `whitepaper/bounded_perps_model.py`.

Run:

```
python3 packages/perps/whitepaper/bounded_perps_model.py \
--fetch \
--output-dir packages/perps/whitepaper/generated

cd packages/perps/whitepaper
python3 -m unittest -v test_bounded_perps_model.py

cd ../../..
python3 packages/perps/whitepaper/render_whitepaper.py
```

Machine-readable results are in `whitepaper/generated/results.json`, and the derived observation series is `whitepaper/generated/ecb_plether_basket.csv`. The publication renderer writes `output/pdf/plether-perps-bounded-credit-whitepaper.pdf`. The reproducibility guide describes the fixed data window, checksum behavior, and limitations.

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